

In-situ Ultrasound Guided Closed Long Bone Shaft Fracture Treatments

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Figure 1: (a) Surgical scene of in-situ ultrasound guided closed long bone shaft fracture treatment. (b) Identification and marking of anatomical landmarks for entry point determination (red arrow). (c) AR ultrasound-guided guidewire insertion (red arrow). (d, e) AR ultrasound-guided fracture reduction (yellow and blue arrows represent proximal and distal segments, respectively).

ABSTRACT

Closed Reduction and Internal Fixation (CRIF) is the preferred treatment for closed long bone shaft fractures, where frequent X-rays imaging reduce efficiency and increase radiation exposure. Intra-operative ultrasound offers real-time, radiation-free bone imaging, but challenges orthopedic surgeons. To address this, we developed an Augmented Reality (AR) navigation system with in-situ ultrasound visualization, combining on-device infrared tool tracking via an AR headset's depth sensor with remote rendering for accurate, low-latency images. A standard operating procedure for surgical application was also proposed. Clinical trials with six patients showed significant reduction in fluoroscopy dose and operation time.

Index Terms: Augmented Reality, In-situ Ultrasound, Closed Fracture Reduction, Clinical Validation.

1 INTRODUCTION

Closed Reduction and Internal Fixation (CRIF) is the preferred minimally invasive treatment for long bone shaft fractures but heavily relies on intraoperative X-rays for fracture visualization, increasing radiation exposure risks. Intraoperative ultrasound offers a radiation-free, real-time imaging alternative but is limited to specific imaging planes, requiring dynamic probe manipulation and multi-dimensional data integration on 2D screens, complicating operations [5]. Augmented Reality (AR) enhances spatial information by overlaying virtual ultrasound images onto actual scanning sites, linking reduction actions with ultrasound image changes. However, current in-situ ultrasound navigation research focuses on display accuracy [3] and data streaming [1], neglecting clinical considerations, thereby hindering stable surgical application. Additionally, existing AR ultrasound systems are mainly used for biopsies [1, 2] and vascular access [3, 4]. To address these issues, we developed an AR navigation system using on-device infrared tool tracking and re-

mote rendering to ensure performance and tracking accuracy during CRIF. We also proposed a standard CRIF operation pipeline based on this system for implant entry point determination and fracture reduction. Clinical trials, including a control group with traditional X-ray guidance, demonstrated the feasibility and effectiveness of our approach.

2 IN-SITU ULTRASOUND NAVIGATION

To achieve high-precision, low-latency in-situ ultrasound visualization, we split the system design into an AR headset and a personal computer (PC), wirelessly connected via network protocols. The E2 Portable Color Doppler Ultrasound System (SonoScape, Shenzhen, China) and linear probe (L741) provided real-time ultrasound images, which were transmitted to the PC for rendering through an integrated image acquisition card. The rendered images were then streamed to the HoloLens 2 for 3D visualization using a remote rendering protocol. Additionally, a tracking case was designed and 3D printed to rigidly fix the retro-reflective markers, which mechanically cooperated with the ultrasound probe. The headset's built-in depth sensor determined the marker poses for real-time localization and tracking of the ultrasound probe, incorporating adaptive depth error correction to enhance tracking stability. Position data were synchronized to the PC via UDP protocol, providing real-time pose information for rendering. This method transmits only visualization-resolution images, reducing redundant data transmission and minimizing system latency, achieving additional system delay of 16.22 ± 11.45 ms.

3 ULTRASOUND-GUIDED CRIF OPERATIONS

We proposed a standard pipeline that integrates in-situ ultrasound into multiple steps of the CRIF procedure to maximize system efficiency, taking intramedullary nailing (IMN) as an example. During entry point determination, the wide-range scan was first performed by moving the ultrasound probe perpendicular to the image plane around the proximal bone to identify the entry point using anatomical landmarks. The probe was then aligned with the long bone shaft near the entry point, and a guidewire was inserted into the bone using in-plane manipulation. To ensure correct guidewire direction, the probe was rotated to the transverse plane and moved to the distal end of the proximal bone segment. Observing the bone contour in the ultrasound images, surgeons can dynamically adjust and advance the guidewire toward the fracture line. During fracture reduction, the ultrasound probe was first placed perpendicular to

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Table 1: Evaluation metrics of the clinical experiment

Group	Number	N_E (times)	N_R (times)	N_S (times)	T_E (minutes)	T_R (minutes)	T_S (minutes)
Experimental Group	1	2	2	41	3.550	4.500	51.733
	2	2	2	67	4.583	4.667	74.583
	3	4	2	94	4.750	4.750	74.400
	Mean(Std)	2.667(1.155)	2(0)	67.333(26.502)	4.294(0.650)	4.639(0.127)	66.905(13.140)
Control Group	1	28	107	214	10.350	17.250	72.183
	2	15	41	232	6.533	5.533	116.867
	3	35	149	270	12.150	50.433	129.100
	Mean(Std)	26(10.149)	99(54.443)	238.667(28.589)	9.678(2.868)	24.405(23.290)	106.050(29.961)

the bone axis and moved from proximal to distal to locate the fracture line. It was then aligned with the bone axis to visualize both sides of the fractured bone. During traction reduction, the probe remained relatively stationary at the fracture site, allowing continuous visualization of bone movement throughout the reduction process.

4 CLINICAL EXPERIMENT

This study involved six patients undergoing closed intramedullary nailing, randomly assigned to experimental and control groups. We assessed the fluoroscopic exposures (N_E , N_R , N_S) and duration (T_E , T_R , T_S) for entry point determination (E), fracture reduction (R) and the entire surgery (S). The control group received standard X-ray guidance procedures. The experimental group followed the proposed AR ultrasound-navigation method for entry point determination and fracture reduction, while using X-ray for step-end confirmations or whenever additional spatial information was needed. This study was approved as a pilot experiment for sample size estimation during IRB application by the Beijing Tsinghua Changgung Hospital Medical Ethics Committee (Approval No.: 24204-4-02).

5 RESULTS

For the experimental group, the fluoroscopy times N_E , N_R and N_S were 2.667 ± 1.155 , 2.000 ± 0.000 , and 67.333 ± 26.502 . In contrast, fluoroscopy times in control group reached 26 ± 10.149 , 99 ± 54.443 , and 238.667 ± 28.589 . The adoption of AR ultrasound navigation significantly reduced the exposure times by 89.742% and 97.980% in entry point determination and bone fracture reduction. Meanwhile, the exposure times during the whole CRIF procedure was reduced by 71.788%. Furthermore, the mean times of T_E , T_R and T_S in the experimental group were 4.294 ± 0.650 , 4.639 ± 0.127 , and 66.905 ± 13.140 minutes, respectively. Comparatively, 9.678 ± 2.868 , 24.405 ± 23.290 , and 106.050 ± 29.961 minutes were taken for the control group. The time reduction for entry point determination and bone fracture reduction reached 55.631% and 80.992%, while the total surgery time was reduced by 36.912%.

6 DISCUSSION

For entry point determination in CRIF, the main challenge of ultrasound guidance is accurately identifying the three-dimensional entry point using two orthogonal images. In-situ display enhances spatial information, enabling the mapping of anatomical landmarks from ultrasound images to the real patient space and reducing the need to consciously reconcile the ultrasound screen with the probe. Once the entry point is established, maintaining the guidewire within the ultrasound plane allows real-time assessment of its spatial relationship with the target bone, avoiding multiple exploratory adjustments required with X-ray navigation. During fracture reduction, X-ray guidance has low spatial efficiency due to the lack of depth information. Although ultrasound provides only specific planar spatial information, its real-time imaging and the spatial information from in-situ display enable comprehensive and effective assessment of fracture status through continuous scanning.

Compared to previous studies using ultrasound alone for fracture reduction [5], AR in-situ visualization reduced fluoroscopy times from 20 to 2 and operation time from 16 minutes to under 4.5 minutes. While the sample size is limited, further clinical trials are needed to validate the system's reliability. As ultrasound cannot penetrate cortical bone and only provides surface spatial information, X-ray usage remains essential for assessing the relative positions of surgical tools and implants with the bone. By extracting and reconstructing general bone surfaces through rapid scanning, the system could further reduce the expertise required for ultrasound operation and interpretation, facilitating its routine application in long bone fracture treatment.

7 CONCLUSION

This paper introduced an AR ultrasound guidance system for long bone shaft fracture reduction, integrating on-device target tracking and remote rendering to enhance performance. We proposed a standard clinical integration pipeline and validated the system's feasibility and effectiveness through clinical experiments. Compared to traditional X-ray navigation, AR ultrasound guidance reduced fluoroscopy times for entry point determination and fracture reduction by 89.742% and 97.980%, respectively, and reduced the corresponding operation times by 55.631% and 80.992%. These reductions minimize radiation exposure risks for medical staff and patients while maintaining high-quality treatment.

ACKNOWLEDGMENTS

This research was supported by NSFC(U22A20355, U20A20389), National Key R&D Program of China (2022YFC2405304) and Medivis. Inc.

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